Nourin Abdelhafiz

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EDUCATION

Toronto Metropolitan University
Bachelor of Engineering in Computer Engineering — Software Engineering Focus
GPA: **3.39/4.33** | 2026

WORK EXPERIENCE

Full-Stack Web Developer — Educational Management System

Freeance Project

- Designed and developed a **role-based educational management system** for teachers, teaching assistants, administrators, and students using **Java, Spring Boot, JSP, and PostgreSQL**.
- Implemented **authentication, session management, role-based permissions, and database operations** across academic, fluency, writing, and administrative modules.
- Designed and executed **functional, smoke, integration, regression, usability, and end-to-end tests** for login, student placement, grading, permissions, and account-management workflows.
- Developed **Selenium WebDriver automation scripts** for critical browser workflows and used **JUnit and Mockito** to validate backend services, application logic, and expected system behaviour.
- Validated application endpoints and **CRUD operations** using **Postman, SQL queries, browser developer tools, logs, screenshots, and structured defect documentation** to identify and resolve issues.

Junior Full-Stack Developer Intern

ROAA Construction

- Developed responsive frontend components by translating UI/UX requirements into maintainable **HTML, CSS, and JavaScript** functionality.
- Integrated frontend pages with backend services through **REST APIs**, supporting dynamic data exchange and client-side application logic.
- Performed **functional, usability, and cross-browser testing** to verify navigation, forms, responsive layouts, and consistent application behaviour.
- Investigated frontend and integration issues using **browser developer tools, console logs, API responses, and debugging techniques**.
- Collaborated with developers and stakeholders to clarify requirements, resolve defects, and improve features through **iterative development and Git-based version control**.

PROJECTS

AI-Enhanced Educational Game for Children with Learning Disabilities

Unity, C#, Q-Learning, JSON, Git

- Developed an interactive educational game supporting children with **dyslexia, dyscalculia, and auditory processing challenges**.
- Integrated a **Q-learning model** that evaluated player performance and adjusted question difficulty using mistakes, completion time, hints, and audio usage.

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- Created structured test scenarios to validate **gameplay logic, difficulty selection, scoring, data collection, and user-interface behaviour**.
- Analyzed test results across different simulated player profiles to identify unexpected decisions and improve the reliability of the learning model.

Smart Parking System — Distributed Systems Project

React, JavaScript, Python, FastAPI, REST APIs

- Developed a distributed parking application that allowed users to view availability, reserve parking spaces, and receive real-time status updates.
- Built **FastAPI microservices** for reservations, parking status, sensor events, and dynamic pricing, with a React-based user interface.
- Designed and tested **REST API endpoints, service integrations, error responses, and reservation workflows** using Postman and structured test data.
- Simulated IoT sensor events and performed **integration and regression testing** to verify that availability changes were reflected correctly across the system.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C#, SQL
Quality Engineering: Selenium WebDriver, JUnit, Mockito, Postman, test-case design, functional testing, smoke testing, regression testing, integration testing, API testing, end-to-end testing, defect documentation
Frontend: React, JSP, HTML5, CSS3, Tailwind CSS, responsive web development
Backend: Spring Boot, FastAPI, REST APIs, microservices, object-oriented programming
Databases: PostgreSQL, MySQL, database validation, CRUD testing
Tools and Practices: Git, Docker, IntelliJ IDEA, browser developer tools, debugging, CI/CD fundamentals, Agile development fundamentals, AWS fundamentals